WhiteBox Testing

TV DESA Kota Batu

Version	By	Description
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Tugas Pre-Prakktikum

1. **Contoh Pengerjaan menggunakan Statement Coverage**

$$Statement\ coverage = \frac{Number\ of\ executed\ statements}{Total\ number\ of\ statements} \times 100\%$$

- **Login Aplikasi**

```php
public function login(Request $request)
{
    $credentials = $request->validate([
        'email' => 'required|string|email',
        'password' => 'required|string',
    ]);
    if (Auth::attempt(array_merge($credentials, ['role' => 'visitor']))) {
        $request->session()->regenerate();
        return response()->json(['message' => 'Visitor logged in successfully.'], 200);
    }
    return response()->json(['message' => 'Invalid credentials.'], 401);
}
```

Scenario 1 :
Jika melakukan verifikasi login dengan data yang valid

$$Statement\ coverage = \frac{11}{12} \times 100\% = 91{,}6\%$$

Scenario 2 :
Jika melakukan verifikasi login dengan salah satu data yang kosong

$$Statement\ coverage = \frac{8}{12} \times 100\% = 66{,}6\%$$

Scenario 3 :
Jika melakukan verifikasi login dengan data yang tidak valid

$$Statement\ coverage = \frac{10}{12} \times 100\% = 83{,}3\%$$

TC ID#	Input	Output	Statement Coverage	Status
TC-1	Mengecek verifikasi login dengan data yang valid	Menampilkan notifikasi “Login successful”	$\frac{11}{12} \times 100\% = 91{,}6\%$	Berhasil
TC-2	Mengecek verifikasi login dengan salah satu data yang kosong	Menampilkan notifikasi “An error occured, please try again”	$\frac{8}{12} \times 100\% = 66{,}6\%$	Berhasil
TC-3	Mengecek verifikasi login dengan data yang tidak valid	Menampilkan notifikasi “Invalid Credentials”	$\frac{10}{12} \times 100\% = 83{,}3\%$	Berhasil

```php
public function register(Request $request)
    {
        try {
            $validatedData = $request->validate([
                'name' => 'required|string|max:255',
                'email' => 'required|string|email|max:255|unique:users',
                'password' => 'required|string|min:8',
            ]);
            $user = User::create([
                'name' => $request->name,
                'email' => $request->email,
                'password' => Hash::make($request->password),
                'role' => 'visitor',
            ]);
            Auth::login($user);
            return redirect()->route('home');
        } catch (\Exception $e) {
            return redirect()->back()->withErrors(['error' => $e->getMessage()]);
        }
    }
```

- **Register**

Scenario 1 :
Melakukan register dengan data yang valid

$$Statement\ coverage = \frac{18}{20} \times 100\% = 90\%$$

Scenario 2 :

Melakukan register dengan data tidak valid

$$Statement\ coverage = \frac{12}{20} \times 100\% = 60\%$$

Scenario 3 :

Melakukan register dengan email yang sudah terdaftar

$$Statement\ coverage = \frac{12}{20} \times 100\% = 60\%$$

TC ID#	Input	Output	Statement Coverage	Status
TC-4	Mengecek verifikasi register dengan data yang valid	Berhasil login dan di arahkan ke homepage	$\frac{18}{20} \times 100\% = 90\%$	Berhasil
TC-5	Verifikasi register dengan data tidak valid	Gagal login, muncul pesan error	$\frac{12}{20} \times 100\% = 60\%$	Berhasil
TC-6	Verifikasi register dengan email sudah terdaftar	Gagal login, muncul pesan error	$\frac{12}{20} \times 100\% = 60\%$	Berhasil

- **Tambah Berita**

```php
public function store(Request $request)
    {
        \Illuminate\Support\Facades\Log::info('Store method called');
        \Illuminate\Support\Facades\Log::info($request->all());
        $validatedData = $request->validate([
            'title' => 'required|max:255',
            'date' => now(),
            'description' => 'required',
            'teaser' => 'required',
            'image' => 'nullable|image|mimes:jpeg,png,jpg,gif|max:2048',
            'author' => 'required|max:255',
            'category' => 'required',
            'status' => 'nullable',
        ]);
        if ($request->hasFile('image')) {
            $imagePath = $request->file('image')->store('berita_images', 'public');
            $validatedData['image'] = $imagePath;
        }
        if ($validatedData['category'] === 'CitizenJournalist') {
            $validatedData['status'] = 'pending';
        } else {
            $validatedData['status'] = 'verified';
        }
        $berita = Berita::create($validatedData);
        if (strtolower($validatedData['category']) === 'CitizenJournalist') {
            $redirectUrl = '/citizen';
        } else {
            $redirectUrl = '/admin/' . strtolower($validatedData['category']);
        }
        return redirect($redirectUrl)->with('success', 'Berita created successfully');
    }
```

Scenario 1 :
Menambahkan berita dengan data valid

$$Statement\ coverage = \frac{20}{31} \times 100\% = 64{,}5\%$$

Scenario 2 :
Menambahkan berita dengan form gambar kosong

$$Statement\ coverage = \frac{20}{31} \times 100\% = 64{,}5\%$$

Scenario 3 :

Menambahkan berita dengan data invalid (salah satu form kosong)

$$Statement\ coverage = \frac{16}{31} \times 100\% = 51{,}6\%$$

TC ID#	Input	Output	Statement Coverage	Status
TC-7	Menambah berita dengan data valid	Berita berhasil ditambahkan	$\frac{20}{31} \times 100\% = 64{,}5\%$	Berhasil
TC-8	Menambah berita dengan form image kosong	Berita gagal ditambahkan	$\frac{20}{31} \times 100\% = 64{,}5\%$	Berhasil
TC-9	Menambah berita dengan data invalid	Berita berhasil ditambahkan	$\frac{16}{31} \times 100\% = 51{,}6\%$	Berhasil

- Edit berita

```php
public function update(Request $request, $id)
    {
        $berita = Berita::findOrFail($id);
        $validatedData = $request->validate([
            'title' => 'required|max:255',
            'description' => 'required',
            'image' => 'nullable|image|mimes:jpeg,png,jpg,gif|max:2048',
            'category' => 'required',
            'author' => 'required|max:255',
        ]);
        if ($request->hasFile('image')) {
            $imagePath = $request->file('image')->store('berita_images', 'public');
            $validatedData['image'] = $imagePath;
        }
        $berita->update($validatedData);
        $redirectUrl = '/admin/' . strtolower($validatedData['category']);
        return redirect($redirectUrl)->with('success', 'Berita created successfully');
    }
```

Scenario 1 :

Merubah berita dengan data valid

$$Statement\ coverage = \frac{18}{18} \times 100\% = 100\%$$

Scenario 2 :

Merubah berita dengan image kosong

$$Statement\ coverage = \frac{11}{18} \times 100\% = 61{,}1\%$$

Scenario 3 :

Merubah berita dengan data invalid (salah satu form kosong/melebihi batas inputan)

$$Statement\ coverage = \frac{11}{18} \times 100\% = 61{,}1\%$$

Tugas Praktikum

2. **Teknik White Box Branch Coverage**

- **Login**

```php
public function login(Request $request)
    {
        $credentials = $request->validate([
            'email' => 'string|email',
            'password' => 'string',
        ], ['email.required' => 'Email field cannot be empty']);

        if (Auth::attempt(array_merge($credentials, ['role' => 'visitor']))) {
            $request->session()->regenerate();
            return redirect()->route('home');
        }
        return back()->withErrors([
            'email' => 'The provided credentials do not match our records.',
        ]);
    }
```

Skenario 1 : Validasi Email

1. Email Valid
2. Email Kosong
3. Email invalid

$$Branch\ Coverage = \frac{3}{3} \times 100\% = 100\%$$

Skenario 2 : Validasi Password

1. Password Valid
2. Password Kosong
3. Password invalid

$$Branch\ Coverage = \frac{3}{3} \times 100\% = 100\%$$

Skenario 3 : Autntikasi

1. Autentikasi Berhasil
2. Autentikasi Gagal

$$Branch\ Coverage = \frac{2}{2} \times 100\% = 100\%$$

Total Cabang : 8

Skenario	Covered Test Case	Branch Coverage
Skenario 1	TC3	$\frac{3}{7} \times 100\% = 42.8\%$
Skenario 2	TC4	$\frac{3}{7} \times 100\% = 42.8\%$
Skenario 3	TC1, TC2	$\frac{2}{7} \times 100\% = 28.55\%$

- Register

```php
public function register(Request $request)
    {
        try {
            $validatedData = $request->validate([
                'name' => 'required|string|max:255',
                'email' => 'required|string|email|max:255|unique:users',
                'password' => 'required|string|min:8',
            ]);
            $user = User::create([
                'name' => $request->name,
                'email' => $request->email,
                'password' => Hash::make($request->password),
                'role' => 'visitor',
            ]);
            Auth::login($user);
            return redirect()->route('home');
        } catch (\Exception $e) {
            return redirect()->back()->withErrors(['error' => $e->getMessage()]);
        }
    }
```

Skenario 1 : Validasi username (2 cabang)

a. Username kosong
b. Panjang username < 255

$$Branch\ Coverage = \frac{2}{2} \times 100\% = 100\%$$

Skenario 2 : Validasi password (2 cabang)

a. Password kosong
b. Panjang password > 8

$$Branch\ Coverage = \frac{2}{2} \times 100\% = 100\%$$

Skenario 3 : Validasi email (4 cabang)

a. Email kosong
b. Email valid
c. Email <255
d. Email unik

$$Branch\ Coverage = \frac{4}{4} \times 100\% = 100\%$$

Skenario 4 : Autentikasi Registrasi (2 cabang)

a. Autentikasi berhasil
b. Autentikasi gagal

$$Branch\ Coverage = \frac{2}{2} \times 100\% = 100\%$$

Total Cabang (Branches) = 10

Skenario	Covered Test Case	Branch Coverage
Skenario 1	TC3	$\frac{2}{10} \times 100\% = 20\%$
Skenario 2	TC2	$\frac{2}{10} \times 100\% = 20\%$
Skenario 3	TC1	$\frac{4}{10} \times 100\% = 40\%$
Skenario 4	TC4	$\frac{2}{10} \times 100\% = 20\%$

- Tambah berita

```php
public function store(Request $request)
    {
        \Illuminate\Support\Facades\Log::info('Store method called');
        \Illuminate\Support\Facades\Log::info($request->all());
        $validatedData = $request->validate([
            'title' => 'required|max:255',
            'date' => now(),
            'description' => 'required',
            'teaser' => 'required',
            'image' => 'nullable|image|mimes:jpeg,png,jpg,gif|max:2048',
            'author' => 'required|max:255',
            'category' => 'required',
            'status' => 'nullable',
        ]);
        if ($request->hasFile('image')) {
            $imagePath = $request->file('image')->store('berita_images', 'public');
            $validatedData['image'] = $imagePath;
        }
        if ($validatedData['category'] === 'CitizenJournalist') {
            $validatedData['status'] = 'pending';
        } else {
            $validatedData['status'] = 'verified';
        }
        $berita = Berita::create($validatedData);
        if (strtolower($validatedData['category']) === 'CitizenJournalist') {
            $redirectUrl = '/citizen';
        } else {
            $redirectUrl = '/admin/' . strtolower($validatedData['category']);
        }
        return redirect($redirectUrl)->with('success', 'Berita created successfully');
    }
```

Skenario 1: Validasi Data (12 cabang)

a. Judul terisi
b. Judul < 255
c. Deskripsi terisi
d. Deskripsi >50
e. Teaser terisi
f. Teaser < 255
g. Image terisi
h. Image format sesuai
i. Image < 2MB
j. Author terisi

k. Author < 255
l. Kategori terisi

$$Branch\ Coverage = \frac{12}{12} \times 100\% = 100\%$$

Total Cabang (Branches) =

Skenario	Covered Test Case	Branch Coverage
Skenario 1	TC15 & TC16	$\frac{12}{12} \times 100\% = 100\%$

- Edit Berita

```php
public function update(Request $request, $id)
    {
        $berita = Berita::findOrFail($id);
        $validatedData = $request->validate([
            'title' => 'required|max:255',
            'description' => 'required',
            'image' => 'nullable|image|mimes:jpeg,png,jpg,gif|max:2048',
            'category' => 'required',
            'author' => 'required|max:255',
        ]);
        if ($request->hasFile('image')) {
            $imagePath = $request->file('image')->store('berita_images', 'public');
            $validatedData['image'] = $imagePath;
        }
        $berita->update($validatedData);
        $redirectUrl = '/admin/' . strtolower($validatedData['category']);
        return redirect($redirectUrl)->with('success', 'Berita created successfully');
    }
```

Skenario 1: Validasi Data Baru (12 cabang)
a. Judul terisi
b. Judul < 255
c. Deskripsi terisi
d. Deskripsi >50
e. Teaser terisi
f. Teaser < 255
g. Image terisi
h. Image format sesuai

i. Image < 2MB
j. Author terisi
k. Author < 255
l. Kategori terisi

$$Branch\ Coverage = \frac{12}{12} \times 100\% = 100\%$$

Total Cabang (Branches) = 14

Skenario	Covered Test Case	Branch Coverage
Skenario 1	TC15 & TC16	$\frac{14}{14} \times 100\% = 100\%$

2. **Teknik White Box Decision Coverage**

$$Decision\ Coverage = \frac{Number\ of\ decision\ outcomes\ exercised}{Total\ number\ of\ decision\ outcomes} \times 100\%$$

- **Login**

```php
public function login(Request $request)
    {
        $credentials = $request->validate([
            'email' => 'string|email',
            'password' => 'string',
        ], ['email.required' => 'Email field cannot be empty']);

        if (Auth::attempt(array_merge($credentials, ['role' => 'visitor']))) {
            $request->session()->regenerate();
            return redirect()->route('home');
        }
        return back()->withErrors([
            'email' => 'The provided credentials do not match our records.',
        ]);
    }
```

Skenario 1 : Validasi email (2 pilihan)
a. Email valid (true)

b. Email invalid (false)

$$Decision\ Coverage = \frac{2}{2} \times 100\% = 100\%$$

Skenario 2 : Validasi password (2 pilihan)

a. Password valid (true)
b. Password invalid (false)

$$Decision\ Coverage = \frac{2}{2} \times 100\% = 100\%$$

Skenario 3 : Autentikasi (2 pilihan)

a. Autentikasi berhasil (true)
b. Autentikasi gagal (false)

$$Decision\ Coverage = \frac{2}{2} \times 100\% = 100\%$$

Total Pilihan (Decisions) = 6

Skenario	Covered Test Case	Decision Coverage
Skenario 1	TC3	$\frac{2}{6} \times 100\% = 33.33\%$
Skenario 2	TC2	$\frac{2}{6} \times 100\% = 33.33\%$
Skenario 3	TC1 & TC4	$\frac{2}{6} \times 100\% = 33.33\%$

- **Register**

```php
public function register(Request $request)
    {
        try {
            $validatedData = $request->validate([
                'name' => 'required|string|max:255',
                'email' => 'required|string|email|max:255|unique:users',
                'password' => 'required|string|min:8',
            ]);
            $user = User::create([
                'name' => $request->name,
                'email' => $request->email,
                'password' => Hash::make($request->password),
                'role' => 'visitor',
            ]);
            Auth::login($user);
            return redirect()->route('home');
        } catch (\Exception $e) {
            return redirect()->back()->withErrors(['error' => $e->getMessage()]);
        }
    }
```

Skenario 1 : Validasi username

- Panjang username <= 255 (true/false)

- Username kosong ? (true/false)

$$Decision\ Coverage = \frac{4}{4} \times 100\% = 100\%$$

Skenario 2 : Validasi email

- Email kosong (true/false)
- Email sesuai format (true/false)
- Panjang email <=255(true/false)
- Email unik (true/false)

$$Decision\ Coverage = \frac{8}{8} \times 100\% = 100\%$$

Skenario 3 : Validasi password

- Panjang password >= 255 ? (true/false)
- password kosong ? (true/false)

$$Decision\ Coverage = \frac{4}{4} \times 100\% = 100\%$$

Total Pilihan (Decisions) = 16

Skenario	Covered Test Case	Decision Coverage
Skenario 1	T3	$\frac{10}{16} \times 100\% = 62{,}5\%$
Skenario 2	T2	$\frac{8}{16} \times 100\% = 50\%$
Skenario 3	T1 & T4	$\frac{2}{16} \times 100\% = 12{,}5\%$

- **Tambah Berita**

```php
public function store(Request $request)
    {
        \Illuminate\Support\Facades\Log::info('Store method called');
        \Illuminate\Support\Facades\Log::info($request->all());
        $validatedData = $request->validate([
            'title' => 'required|max:255',
            'date' => now(),
            'description' => 'required',
            'teaser' => 'required',
            'image' => 'nullable|image|mimes:jpeg,png,jpg,gif|max:2048',
            'author' => 'required|max:255',
            'category' => 'required',
            'status' => 'nullable',
        ]);
        if ($request->hasFile('image')) {
            $imagePath = $request->file('image')->store('berita_images', 'public');
            $validatedData['image'] = $imagePath;
        }
        if ($validatedData['category'] === 'CitizenJournalist') {
            $validatedData['status'] = 'pending';
        } else {
            $validatedData['status'] = 'verified';
        }
        $berita = Berita::create($validatedData);
        if (strtolower($validatedData['category']) === 'CitizenJournalist') {
            $redirectUrl = '/citizen';
        } else {
            $redirectUrl = '/admin/' . strtolower($validatedData['category']);
        }
        return redirect($redirectUrl)->with('success', 'Berita created successfully');
    }
```

Skenario 1 : Validasi judul

- Panjang judul <= 255 (true/false)
- judul kosong (true/false)

$$Decision\ Coverage = \frac{4}{4} \times 100\% = 100\%$$

Skenario 2 : Validasi deskripsi

- Panjang deskripsi > 50 (true/false)
- deskripsi kosong (true/false)

$$Decision\ Coverage = \frac{4}{4} \times 100\% = 100\%$$

Skenario 3 : Validasi teaser

- Panjang judul <= 255 (true/false)
- Panjang teaser >10 (true/false)
- judul kosong (true/false)

$$Decision\ Coverage = \frac{6}{6} \times 100\% = 100\%$$

Skenario 4 : Validasi image

- image kosong (true/false)
- file image sesuai format (true/false)
- Size image < 2MB (true/false)

$$Decision\ Coverage = \frac{6}{6} \times 100\% = 100\%$$

Skenario 5 : Validasi author

- author kosong (true/false)
- panjang author <255 (true/false)

$$Decision\ Coverage = \frac{4}{4} \times 100\% = 100\%$$

Skenario 6 : Validasi categori

- kategori kosong (true/false)

$$Decision\ Coverage = \frac{2}{2} \times 100\% = 100\%$$

Total Pilihan (Decisions) = 26

Skenario	Covered Test Case	Decision Coverage
Skenario 1	TC15 & TC16	$\frac{4}{26} \times 100\% = 15.3\%$
Skenario 2	TC15 & TC16	$\frac{4}{26} \times 100\% = 15.3\%$
Skenario 3	TC15 & TC16	$\frac{6}{26} \times 100\% = 23\%$
Skenario 4	TC15 & TC16	$\frac{6}{26} \times 100\% = 23\%$
Skenario 5	TC15 & TC16	$\frac{4}{26} \times 100\% = 15.3\%$
Skenario 6	TC15 & TC16	$\frac{2}{26} \times 100\% = 7.6\%$

- **Edit Berita**

```php
public function update(Request $request, $id)
    {
        $berita = Berita::findOrFail($id);
        $validatedData = $request->validate([
            'title' => 'required|max:255',
            'description' => 'required',
            'image' => 'nullable|image|mimes:jpeg,png,jpg,gif|max:2048',
            'category' => 'required',
            'author' => 'required|max:255',
        ]);
        if ($request->hasFile('image')) {
            $imagePath = $request->file('image')->store('berita_images', 'public');
            $validatedData['image'] = $imagePath;
        }
        $berita->update($validatedData);
        $redirectUrl = '/admin/' . strtolower($validatedData['category']);
        return redirect($redirectUrl)->with('success', 'Berita created successfully');
    }
```

Skenario 1 : Validasi judul baru

- Panjang judul <= 255 (true/false)
- judul kosong (true/false)

$$Decision\ Coverage = \frac{4}{4} \times 100\% = 100\%$$

Skenario 2 : Validasi deskripsi baru

- Panjang deskripsi > 50 (true/false)
- deskripsi kosong (true/false)

$$Decision\ Coverage = \frac{4}{4} \times 100\% = 100\%$$

Skenario 3 : Validasi teaser baru

- Panjang judul <= 255 (true/false)
- Panjang teaser >10 (true/false)
- judul kosong (true/false)

$$Decision\ Coverage = \frac{6}{6} \times 100\% = 100\%$$

Skenario 4 : Validasi image baru

- image kosong (true/false)
- file image sesuai format (true/false)
- Size image < 2MB (true/false)

$$Decision\ Coverage = \frac{6}{6} \times 100\% = 100\%$$

Skenario 5 : Validasi author baru

- author kosong (true/false)
- panjang author <255 (true/false)

$$Decision\ Coverage = \frac{4}{4} \times 100\% = 100\%$$

Skenario 6 : Validasi kategori baru

- kategori kosong (true/false)

$$Decision\ Coverage = \frac{2}{2} \times 100\% = 100\%$$

Total Pilihan (Decisions) = 26

Skenario	Covered Test Case	Decision Coverage
Skenario 1	TC15 & TC16	$\frac{4}{26} \times 100\% = 15.3\%$
Skenario 2	TC15 & TC16	$\frac{4}{26} \times 100\% = 15.3\%$
Skenario 3	TC15 & TC16	$\frac{6}{26} \times 100\% = 23\%$
Skenario 4	TC15 & TC16	$\frac{6}{26} \times 100\% = 23\%$
Skenario 5	TC15 & TC16	$\frac{4}{26} \times 100\% = 15.3\%$
Skenario 6	TC15 & TC16	$\frac{2}{26} \times 100\% = 7.6\%$